Technical Assignment: CO_2 Factor Forecasting Using Renewable Energy Data

Objective:
Build a machine learning model that estimates the **CO_2 factor for the next 7 days, with 1 hour resolution,** based on the forecasted **renewable energy production** in the Netherlands.

Background

Nederlandse Energie Data (NED) provides open-access data on energy production in the Netherlands. In this assignment, you will use their data portal to download historical data and develop a predictive model.

Your task is to:

1. Collect and explore historical data on CO_2 emissions and renewable energy production.

2. Build and document a machine learning model that forecasts the **CO_2 factor** for the next 7 days with a resolution of **1 hour**.

3. Evaluate and explain your modeling approach and results.

Data Sources

You must create a free account at NED:
🔗 NED Data Portal Registration

Once registered, download the following datasets:

- **CO_2 Factor (Historical):**
  Totale Elektriciteitsproductie – CO_2 Factor

- **Solar Energy Production (Historical):**
  Zonne-energie

- **Other Renewable Sources (Optional but Recommended):**
  Wind, hydro, or biomass energy production from the same portal.

- **You can use any other data source you may find useful.**

Assignment Requirements

You must:

1. **Data Collection & Preprocessing**
    - Register and obtain access to the NED datasets.
    - Clean, transform, and visualize the relevant datasets (optional).
2. **Modeling**
    - Design a machine learning model (or multiple and compare them) to forecast the **CO_2 factor for the next 7 days, with 1 hour resolution (i.e. 168 values),** using renewable energy production as input features (and any other you find relevant, optional).
    - Justify your choice of algorithms and pre-processing steps.
    - You are free to use any Python library you find most relevant.
3. **Documentation**
    - Clearly explain your methodology, model selection, and results. In a PDF report (max 5 pages, excluding figures).
    - Include a well-documented README with setup instructions and project structure.
4. **Code Quality**
    - Ensure readable, expandable, and commented code.
    - Include any necessary file to reproduce your environment (e.g. `requirements.txt` or `environment.yml)`

Deliverables

Please submit the following with a **link to a GitHub repository**:

- Source code (in **Python**, Notebook or scripts)

- README with setup and a brief explanation
- Report explaining the approach, assumptions, and results (including Visualizations of predictions vs actuals, and error metrics).

Optional:

- Error analysis or feature importance insights